

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years building high-performance distributed systems and real-time data pipelines using Node.js, TypeScript, and cloud-native technologies. Led zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, demonstrating expertise in async programming, messaging systems, and microservices architecture. Experienced with Apache Kafka, Google Pub/Sub, Kubernetes, and Docker for scalable, production-ready solutions handling high-volume transactional workloads. Strong track record of delivering measurable business impact: \$5K/month infrastructure savings and \$40/order revenue features.

CORE COMPETENCIES

Node.js / TypeScript

Distributed Systems

Apache Kafka

Real-time Messaging

Microservices Architecture

Kubernetes / Docker

GCP / AWS / Firebase

MongoDB / PostgreSQL / Firestore

WORK EXPERIENCE

MTailor

Sep 2022 -- Present

Software Engineer (US Remote)

- Engineered distributed messaging pipeline using Google Pub/Sub for real-time bidirectional synchronization between MongoDB and Firestore, enabling zero-downtime migration of 20+ production applications
- Designed async data processing architecture handling parallel production traffic across legacy and modern systems throughout 6-month migration window
- Built 3D visualization feature using video overlay and ffmpeg, directly increasing buyer conversion rates
- Delivered Express Shipping feature end-to-end, generating \$40 additional revenue per order
- Reduced infrastructure costs by \$5,000/month by migrating all services from AWS to GCP with optimized resource allocation
- Wrote SDK documentation and trained ops team on Firebase Dashboard and backup procedures

Dr Wealth

Aug 2021 -- Jan 2022

Software Engineer (Singapore Remote)

- Built latency-optimized real-time data pipelines serving 2M+ Firestore records with current stock market prices from Morningstar APIs
- Developed ad-hoc batch jobs for querying and updating large-scale NoSQL datasets
- Extended backend APIs using Firebase Cloud Functions and Express.js on Heroku
- Added responsive UI components to PWA using React and Tailwind CSS

Mind2matter.co

Jan 2022 -- Jul 2022

Software Engineer (US Remote)

- Built backend services for DeFi applications using Web3 and Node.js
- Delivered production-ready React UIs from Figma designs under tight deadlines

CodeBits

Jan 2020 -- Jul 2021

Software Engineer (Uganda Remote)

- Architected event-driven microservices using Apache Kafka, Docker, and Kubernetes for FIDA Uganda's case management system, enabling independent scaling of distributed nodes
- Led team of 4 developers building legal tech systems for NGOs across Uganda
- Built cross-platform mobile app using React Native and Expo (iOS & Android) with Firebase push notifications
- Developed USSD service communicating with Node.js backend via gRPC for rural legal aid access

PROJECTS

Real-time Pub/Sub Sync Pipeline Production

Bidirectional sync system enabling parallel MongoDB/Firestore operation during live migration with retry logic and dead-letter handling

Node.js, Google Pub/Sub, MongoDB, Firestore, TypeScript

Healthcare WhatsApp Chatbot Production

Integrated OpenAI + Google Vision for prescription OCR via WhatsApp Business API

Node.js, OpenAI API, Google Vision, WhatsApp Business API

Marketplace Recommendation Engine Production

Collaborative + content-based filtering engine in MongoDB for service provider matching

Node.js, MongoDB, Python

EDUCATION

Bachelor of Science in Computer Engineering -- [Makerere University](#)

2018 -- 2022

Kampala, Uganda

SKILLS

Languages: Node.js, TypeScript, Python, Go, JavaScript **Messaging:** Apache Kafka, Google Pub/Sub, RabbitMQ, gRPC

Databases: MongoDB, Firestore, PostgreSQL, SQL **Cloud:** GCP, AWS, Firebase, Heroku **Infrastructure:** Kubernetes, Docker, Linux

Frontend: React, React Native, Tailwind CSS